Replication of a research claim from Popper et al, 2009, in The Leadership Quarterly

Study Information

Hypotheses

H*: A positive relation will be found between openness to experiences and leadership experiences
Statistical test: Structural equation modeling (SEM)
Expected pattern of results: Openness to experiences will predict leadership experiences

Design Plan

Study type

Observational Study - Data is collected from study subjects that are not randomly assigned to a treatment. This includes surveys, “natural experiments,” and regression discontinuity designs.

Blinding

No blinding is involved in this study.

Is there any additional blinding in this study?

As this is an observational study, we will not need to use blinding (i.e., there is not treatment versus control group and researchers will not interact with participants directly).

Study design

This is a replication study of Popper et al. (2009) from the Leadership Quarterly. As such, we attempt to replicate procedures as much as possible from this study.

We follow the same methods of Popper and Amit (2009) with slight modifications to sample and procedures of data gathering. Their all-male sample was drawn from active-duty Israeli military members, which we will not be able to access. Thus, we will use a US sample of men with military experience (may include active duty, reserve, or veterans), which we do not have reason to believe will be substantially different from the original study sample population. Popper and Amit (2009) also used survey administration but provided surveys directly to sample participants while we will obtain our sample using the Prolific online data collection platform in order to reach the desired number of participants meeting our inclusion criteria.

In our replication, we recruit from a large electronic database, Prolific, and set the following criteria: participants must be 1) men, 2) over the age of 18, 3) have experience survey in the U.S. military, and 4) have an Prolific study approval rate of 90% (to inspire quality responses). We are targeting 671 respondents based on the power analysis provided by the SCORE team below.

We are imposing the following custom qualifications on our Prolific recruitment:

1. In Prolific, we have the ability to screen participants with “custom qualifications” - We will only include those designated as having U.S. military experience and study approval rates of 90%. In the Qualtrics survey, we will require participants to respond to a series of questions asking about their military experience (e.g., active versus veteran status, military branch, length of service). If they respond they do NOT have military experience, they are taken to the end of the survey.

2. In Qualtrics, we will also ask participants if they identify as men and are over the age of 18. If they answer no, they are taken to the end of the survey.

Upon inclusion, participants will be directed to an online survey (SURVEY LINK) that includes all the measures assembled from Popper and Amit (2009) that are relevant to testing Hypotheses 4. They will be compensated $1.20 through Prolific upon survey completion.

Measures:

Insufficient information is provided in the Popper and Amit (2009) manuscript to reconstruct the exact scales used for attachment style, openness to experience, and leader experiences. As such, we have tried to contact the authors, but have not yet received response. We will continue to await response but have also constructed measures for the study below based on the information available in the published study.

Attachment Style. Popper and Amit (2009) do not provide the items they used to measure attachment style, but they did reference Hazan and Shaver (1987) and Shaver, Belsky, and Brennan (2000) regarding how they constructed their attachment scale. They also noted they used 13 total items: 6 items to [illegible] anxious dimension and 7 items to represent the avoidant dimension, with lower total scores indicating secure attachment style. As we cannot kn[illegible] were used to construct either scale, we will use the items used in Shaver, Belsky, and Brennan (which were pulled from Collins and Read, 1990). These items include six items for the anxiety scale and twelve items for the two facets of avoidance: closeness and dependence. Items are scored on a 5-point Likert-type

scale from where 1 = not at all characteristic, 2 = somewhat uncharacteristic, 3 = neither characteristic nor uncharacteristic, 4 = somewhat characteristic, and 5 = very characteristic. Higher scores indicate more secure attachment style.

Trait Anxiety. We will measure trait anxiety using the same strategies as Popper and Amit (2009) in using the STAI for adults. Respondents will be provided the following instructions: "Read each statement and then blacken the appropriate circle to the right of the statement to indicate how you feel right now, that is, at this moment. There are no right or wrong answers. Do not spend too much time on any one statement but give the answer which seems to describe your present feelings best." They will then indicate their responses from 1 = Not at all, 2 = Somewhat, 3 = Moderately so, and 4 = Very much so. Higher scores indicate higher trait anxiety.

Openness to Experiences. Similar to the attachment style scale, Popper and Amit (2009) did not supply the items they used for their openness to new experiences scale. They disclosed they used an abbreviated version of the NEO-PI-R openness to experience scale consisting of 24 items. We cannot know the specific items Popper and Amit (2009) used for their study; however, the International Personality Item Pool (IPIP) has 20-item abbreviated NEO Openness to Experience measure that is commonly used in social science research. Thus, we will use this version of the NEO Openness to Experience measure. The following instructions will be given to participants "Describe yourself as you generally are now, not as you wish to be in the future. Describe yourself as you honestly see yourself, in relation to other people you know of the same sex as you are, and roughly your same age. So that you can describe yourself in an honest manner, your responses will be kept in absolute confidence. Indicate for each statement whether it is 1. Very Inaccurate, 2. Moderately Inaccurate, 3. Neither Accurate Nor Inaccurate, 4. Moderately Accurate, or 5. Very Accurate as a description of you." Higher scores indicate higher Openness to Experience.

Leadership Experiences. We will measure leadership experiences using the same strategies as Popper and Amit (2009). The scale used for this construct was created by Popper and Amit (2009) specifically for their study, and was designed to capture the early-life experiences with influence and leadership roles of participants prior to their joining the military. Popper and Amit provided the questions for the scale. As such, participants will be asked to read a set of seven statements about their past leadership experiences and rate each statement regarding the level of experience they have with each. Popper and Amit did not provide instructions for their scale, however, so we created instructions to ensure participants are reflecting on their experiences through high school and prior to joining the military, which is consistent with the desired time period put forth by Popper and Amit for this scale. Participants will see the instructions, " For the following statements, please think about your early life experiences through grade school education and up to high school graduation (i.e., prior to joining the military). Please select the answer below that best describes the extent of experience you had with the topic in each statement during this early life time period." Items will be rated on a scale of 1 to 4 where 1 = Hardly any experience, 2 = Some experience, 3 = Moderate experience, and 4 = Rich experience. Popper and Amit (2009) only suppled the scale anchors at 1 and 4 for this measure, so we had to create the response options for 2 and 3. Higher scores indicate greater extent of early leadership experience.

Other. As we are using a sample of U.S. men with military service, our sample differs from the original study in that 1) participants will not be Israeli, and 2) Participants will not have served in the Israeli military. As such, we will collect the following additional demographic information on our sample so that we may describe and account for the different population:

1. Age
2. Race
3. Highest Education level (Some High School- did not graduate, High school, Bachelor's degree, Graduate school, Post-graduate work)
4. Type of military service (e.g., army, navy, air force, etc.)
5. Length of military service
6. Status of military service (active duty, reserve, or veteran)

Data Analysis
While Popper and Amit (2009) utilized Amos 5 software for their analyses, we will test the hypothesis using Structural Equation Modeling in R Statistical Software. This represents a departure in analysis tool only (i.e., analysis procedures will be the same as the original study) and we have no reason to expect the difference in software to have an effect on results. The original study applied a parceling approach to aggregate latent variable indicators stating, "A rule of thumb is that a measurement model with three indicators per latent construct is ideal, and it can carry up to five indicators without estimation difficulties" (p. 755) and citing Bagozzi and Heatherton (1994). However, substantial advances in computational software abilities have been made since 2009, and R software has no issue estimating models with more than 5 indicators per latent variable and parceling is a controversial practice in structural equation modeling. Thus, we will replicate the random parceling procedure to aggregate latent variable indicators for use in testing the structural model. In addition, we will also run a separate analysis where we include all indicators for each latent variable in our model estimation and provide results for both sets of analyses. Paths will be estimated with Leadership Experience as the dependent variable, Attachment as the independent variable and Trait Anxiety and Openness to Experience as the two mediator variables.

No files selected

Randomization

As this is an observational survey design, there is no randomization (i.e., there are not multiple conditions).

Sampling Plan

Existing Data

Registration prior to creation of data

Explanation of existing data

This project is being preregistered prior to any new data collection.

Data collection procedures

As Popper et al. (2009) used a military sample, we will sample participants with military experience. We are aware of the DoD requirement to have DoD review by the Human Research Protection Officer (HRPO) and we will obtain this HRPO review prior to implementing any aspects of this study.

Participants will be recruited using Prolific online data collection platform. Individuals meeting the study criteria will be able to see our HIT research announcement (abbreviated summary of the study -- see attachment to this application) at which point they may opt to participate in the study. Once potential

participants decide to participate, they will be taken to an online survey (via Qualtrics platform) where they will view the informed consent. Individuals who consent to participate in the research will continue on to the study survey, which contains questions about past attachment experiences (items from Shaver, Belsky, and Brennan, 2000), trait anxiety (STAI for adults), openness to experience (International Personality Item Pool (IPIP) has 20-item abbreviated NEO Openness to Experience measure), and leadership experiences (Scale developed by Popper & Amit, 2009). We must include all of the original scales from the original study as the method of analysis is structural equation modeling, which requires all paths to be estimated to obtain the focal coefficient (relationship between openness to experience and leadership experiences). Participants will also be asked basic demographic questions (e.g., age, racial identity, etc.). Upon completion of the survey, participants will receive a unique code they must supply via Prolific to receive payment ($1.20 per complete response). Questions included in our "screening block" in the survey are not questions that prohibit access to the survey as participants will be screened by Prolific when we specify our qualifications for inclusion (e.g., at least 18 years old, male, living in the US, military experience). These are questions we will use to screen for inclusion in analysis. Any respondents answering they do not meet inclusion criteria will be excluded from analysis.

No files selected

Sample size

The target analytic sample size is 671 participants as determined by COS power analysis.

To achieve the target analytic sample size, we anticipate the need to collect a slightly larger sample size to account for any necessary deletion due to insufficient effort responding or extensive missing data. The target sample size for recruitment is thus 700 participants. Insufficient effort responding will be determined by reviewing time to complete the survey: participants that complete in less than 30 seconds will be removed from the data. Participants missing more than 10% of data per scale will have their responses removed from the data, as will participants who fail both of 2 attention check items.

Sample size rationale

Power calculations were conducted in accordance with the guidelines of the Social Sciences Replication Project (SSRP). However, instead of carrying out two rounds of data collection to achieve the stage 1 sample size target followed by the stage 2 sample size target as in Phase 1 of SCORE, in Phase 2 only one data collection effort will be carried out to achieve a single target analytic sample size.

All replications will aim to achieve the target analytic sample size, which is determined by a power analysis achieving 90% power to detect 50% of the original effect size (formally referred to as the Stage 2 sample size). However, if this sample size is not attainable for the replicating lab, the sample size will instead be calculated by achieving 90% power to detect 75% of the original effect size (formally referred to as the Stage 1 sample size).

For this study, because collecting the Stage 2 sample size would not be feasible under SCORE's allowable budget, the Stage 1 sample size (which is powered to detect 75% of the original effect size) will instead be collected. The Stage 1 sample size for this study is 671 participants.

Documentation of the power analysis conducted for this replication is available at the following view-only link: https://osf.io/q8xpr/?view_only=665f77d6e7ed416fa8877edeaccdf7e3.

Stopping rule

The planned analytic sample size is 671 participants. The target sample size for recruitment is 700 participants. After achieving this sample, sampling will stop and planned analyses will be run.

Variables

Manipulated variables

As this is an observational study, there are no manipulated variables.

No files selected

Measured variables

Variables to be included in this study are described below. As we are using structural equation modeling as the method of analysis, we must capture all study variables previously captured in Popper & Amit (2009) to obtain the target effect size. The independent and dependent variable are labelled as such below:

Attachment (included in the structural model) - We will use the items used in Shaver, Belsky, and Brennan (2000, which were pulled from Collins and Read, 1990). These items include six items for the anxiety scale and twelve items for the two facets of avoidance: closeness and dependence. Items are scored on a 5-point Likert-type scale from where 1 = not at all characteristic, 2 = somewhat uncharacteristic, 3 = neither characteristic nor uncharacteristic, 4 = somewhat characteristic, and 5 = very characteristic. Higher scores indicate more secure attachment style.

Anxiety (Included in the structural model) - We will measure trait anxiety using the same strategies as Popper and Amit (2009) in using the STAI for adults. Respondents will be provided the following instructions: "Read each statement and then blacken the appropriate circle to the right of the statement to indicate how you feel right now, that is, at this moment. There are no right or wrong answers. Do not spend too much time on any one statement but give the answer which seems to describe your present feelings best." They will then indicate their responses from 1 = Not at all, 2 = Somewhat, 3 = Moderately so, and 4 = Very much so. Higher scores indicate higher trait anxiety.

Openness to Experience (independent variable) - The International Personality Item Pool (IPIP) has 20-item abbreviated NEO Openness to Experience measure that is commonly used in social science research. Thus, we will use this version of the NEO Openness to Experience measure. The following instructions will be given to participants "Describe yourself as you generally are now, not as you wish to be in the future. Describe yourself as you honestly see yourself, in relation to other people you know of the same sex as you are, and roughly your same age. So that you can describe yourself in an honest manner, your responses will be kept in absolute confidence. Indicate for each statement whether it is 1. Very Inaccurate, 2. Moderately Inaccurate, 3. Neither Accurate Nor Inaccurate, 4. Moderately Accurate, or 5. Very Accurate as a description of you." Higher scores indicate higher Openness to Experience.

Leadership Experiences (dependent variable)- We will measure leadership experiences using the same strategies as Popper and Amit (2009). The scale used for this construct was created by Popper and Amit (2009) specifically for their study, and was designed to capture the early-life experiences with influence and

leadership roles of participants prior to their joining the military. Popper and Amit provided the questions for the scale. As such, participants will be asked to read a set of seven statements about their past leadership experiences and rate each statement regarding the level of experience they have with each. Popper and Amit did not provide instructions for their scale, however, so we created instructions to ensure participants are reflecting on their experiences through high school and prior to joining the military, which is consistent with the desired time period put forth by Popper and Amit for this scale. Participants will see the instructions, " For the following statements, please think about your early life experiences through grade school education and up to high school graduation (i.e., prior to joining the military). Please select the answer below that best describes the extent of experience you had with the topic in each statement during this early life time period." Items will be rated on a scale of 1 to 4 where 1 = Hardly any experience, 2 = Some experience, 3 = Moderate experience, and 4 = Rich experience. Popper and Amit (2009) only supplied the scale anchors at 1 and 4 for this measure, so we had to create the response options for 2 and 3. Higher scores indicate greater extent of early leadership experience.

Other- As we are using a sample of U.S. men with military service, our sample differs from the original study in that 1) participants will not be Israeli, and 2) Participants will not have served in the Israeli military. As such, we will collect the following additional demographic information on our sample so that we may describe and account for the different population, though these variables will not be included in the hypothesis testing:
Age
Race
Highest Education level (Some High School- did not graduate, High school, Bachelor's degree, Graduate school, Post-graduate work)
Type of military service (e.g., army, navy, air force, etc.)
Length of military service
Status of military service (active duty, reserve, or veteran)

We acknowledge that certain differences in the original study's sample and our proposed sample may result in different observations of effects. For example, it is possible the effect size between openness to experience and leadership experiences will be larger in the American population due to selection effects related to the voluntary nature of American military service (Israeli military service is compulsory). Thus, higher baseline levels of early life leadership experiences may be present for the American sample compared to the Israeli sample, possibly resulting in a stronger observed relationship (i.e., greater than B = .34). Should the effect size for our sample be observed to be substantially higher than that observed in the Popper & Amit (2009) study, we will discuss possible reasons for this in our report. This difference in sample should not interfere with the potential for replicability, however.

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Indices

We are using structural equation modeling to analyze the data and will follow Popper & Amit's strategy for creating 3 random parcels per scale for parameter estimation. Thus, each of the three parcels for each scale (attachment, trait anxiety, openness to experience, and leadership experiences) represents a composite score of the items randomly assigned to that parcel. Further, to run correlations of all study variables, composite scores will be created for each of the four measured variables included in the structural model (attachment, trait anxiety, openness to experience, and leadership experiences)

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Analysis Plan

Statistical models

For the purposes of SCORE, to test H* and attempt to replicate the H* statistical evidence we will use structural equation modeling with randomized parceling, as this was the method used by Popper & Amit (2009). However, we will also test the structural model without parceling the items as a robustness check.

No files selected

Transformations

We will not transform our measured variables.

Inference criteria

For this study, Model 1 was retained as the best fitting model from the Popper & Amit original study and the effect size generated for H4 was B = .34. Thus successful replication should be represented by a statistically significant positive beta coefficient for the relationship between openness to experience and leader experiences.

Data exclusion

If participants indicate they do not meet any of the inclusion criteria (military experience, U.S. resident, male), they will be excluded from analyses.

If participants demonstrate insufficient effort responding as evidenced by marking all items across scales as the same response, they will be excluded from analyses (e.g., the openness to experience scale should vary among given certain items that are reverse-coded).

Participants who finish the survey in less than 30 seconds will be excluded from the data given the length of the survey.

We will also include two attention check items. Participants who fail both attention check items will be removed from the survey

Missing data

If participants are missing substantial responses (more than 60% missing per scale), they will be excluded from analyses. The mean score for all items on a scale will be imputed for missing data given participants completed at least 90% of items per scale.

Exploratory analysis

Not applicable

Other

Other

The original study was conducted with Israeli active-duty male military members. This replication deviates from the original in that we will use U.S. males with military experience (active duty or veteran).

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